

Figure 2. Data records, repository organization and access model

A Archival repository hierarchy

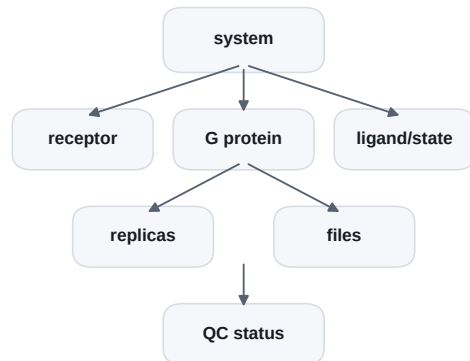
```

archive_root/
manifest/  checksums + version
metadata/  systems + held_back
systems/<system_id>/
  topology/  structures + parameters
  trajectories/rep{1..3}/  full production
  reduced/  aligned strided trajectories
  features/  RMSD, RMSF, contacts
  pocket/  fpocket + GPCRdb mapping
  qc/  frame, PBC, stability reports

code/  scripts + notebooks
web_api_snapshot/  OpenAPI schema

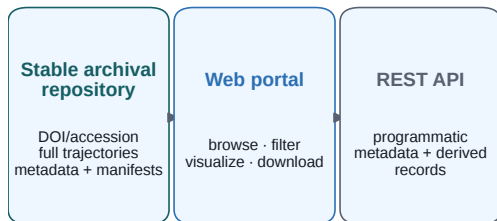
clean v1: 650 archive paths · 2080 portal/API records
    
```

B Metadata schema (per system)



Keys: system_id, UniProt, GPCRdb, class, G-protein family, ligand/source, replica IDs, paths, QC flags, checksums

C Archive-first access model



Portal layer: 2078/2080 record slots present (2 missing); binary checksums pending

D Web portal access layer

